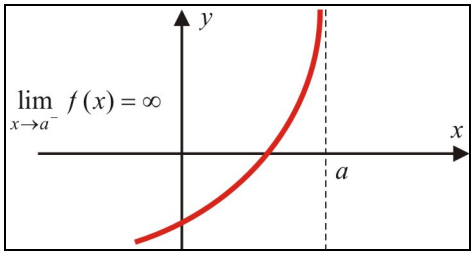
4.3 Vertical and Horizontal Asymptotes

A Vertical Asymptote

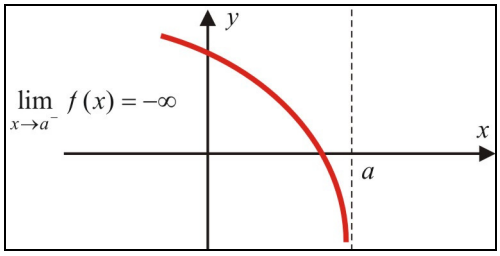
If the value of $f(x)$ can be made *arbitrarily large* by taking x *sufficiently close* to a with $x < a$ then:

$$\lim_{x \to a^-} f(x) = \infty$$

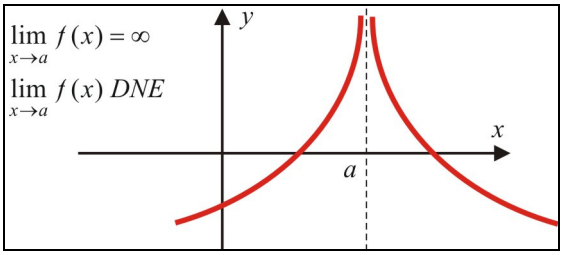
The line $x = a$ is called *vertical asymptote* to the graph of $y = f(x)$.

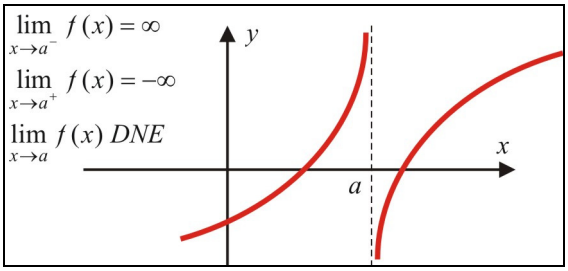


Similarly:



B Notes





C Rational Functions

A rational function of the form $f(x) = \dfrac{p(x)}{q(x)}$ has a vertical asymptote $x = a$ if:

$$q(a) = 0 \text{ and } p(a) \neq 0$$

V.A $x = -3$

$\lim_{x \to -3^-} f(x) = -\infty$

$\lim_{x \to -3^+} f(x) = +\infty$

Ex 1. Find the equation of the vertical asymptote(s) for $y = f(x) = \dfrac{x-2}{x^2-4}$.

$y = \dfrac{x-2}{(x-2)(x+2)}$

$x = -2$

Ex 2. Find the behavior of the function $y = f(x) = \dfrac{-x}{x^2+2x-3}$ near the vertical asymptotes.

$f(x) = \dfrac{-x}{(x+3)(x-1)}$

$\lim_{x \to 1^-} f(x) = +\infty$ V.A $x = 1$

$\lim_{x \to 1^+} f(x) = -\infty$

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D Horizontal Asymptote

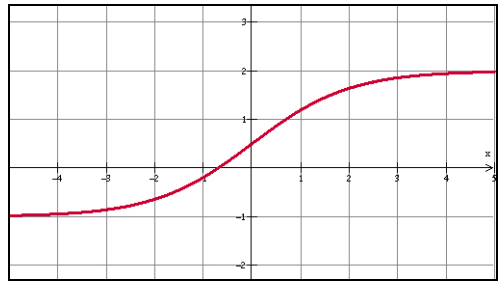
A horizontal line $y = L$ is called horizontal asymptote to the graph of $y = f(x)$ if:

$$\lim_{x\to\infty} f(x) = L \quad or \quad \lim_{x\to-\infty} f(x) = L$$

Notes.

1. A horizontal asymptote may be crossed or touched by the graph of the function.
2. The graph of a function may have at most two horizontal asymptotes (one as $x \to -\infty$ and one as $x \to \infty$) (see the figure on the right).

Ex 3. Find the equation of the horizontal asymptote(s) to the graph of the function $y = f(x)$ represented graphically below.



E Limits to Infinity

If $n \geq 1$, then:

$$\lim_{x\to\pm\infty} x^n = (\pm\infty)^n$$

$$\lim_{x\to\pm\infty} \frac{1}{x^n} = 0$$

Ex 4. Compute each limit.

a) $\lim_{x\to\infty} x =$

b) $\lim_{x\to\infty} x^2 =$

c $\lim_{x\to-\infty} x^3 =$

d) $\lim_{x\to-\infty} x^4 =$

F End behaviour

A polynomial functions

$P(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$

behaves at infinity as the leading term $a_n x^n$.

Ex 5. Compute each limit.

a) $\lim_{x\to\infty} (-3x^4 + 5x^3 - 4)$ $= -\infty$

b) $\lim_{x\to-\infty} (-x^3 - 2x^2 + x)$ $= +\infty$

G Rational Functions

To compute the limits at infinity for a rational function, use the end behavior of the numerator and denominator:

$$\lim_{x\to\pm\infty} \frac{a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0}{b_m x^m + b_{m-1}x^{m-1} + \ldots + b_2 x^2 + b_1 x + b_0} = \lim_{x\to\pm\infty} \frac{a_n x^n}{b_m x^m}$$

Ex 6. Compute each limit.

a) $\lim_{x\to\infty} \frac{-2x^3 + x}{5x^3 + x^2 + 1}$ $= -\frac{2}{5}$

b) $\lim_{x\to-\infty} \frac{3x^2 + 1}{-2x^3 + 3x}$ $= 0$

c) $\lim_{x\to-\infty} \frac{3x^4 + x^2 - x}{-x^2 + x - 1}$ $= -\infty$

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H Horizontal Asymptotes for Rational Functions

A rational function of the form:

$$f(x)=\frac{P_n(x)}{Q_m(x)}=\frac{a_nx^n+a_{n-1}x^{n-1}+\ldots+a_2x^2+a_1x+a_0}{b_mx^m+b_{m-1}x^{m-1}+\ldots+b_2x^2+b_1x+b_0}$$

has

- a horizontal asymptote $y=0$ if $m>n$
- a horizontal asymptote $y=\frac{a_n}{b_m}$ if $m=n$
- no horizontal asymptote if $n>m$

Note: A rational function may have at most one horizontal asymptote.

Ex 7. Find the equation of the horizontal asymptote.

a) $f(x)=\frac{3x^4+1}{-2x^4+3x^2}$ $\quad y=-\frac{3}{2}$

b) $f(x)=\frac{-x^2+2x}{x^3-x^2+2}$ $\quad y=0$

c) $f(x)=\frac{2x^2-3}{x+1}$ $\quad$ DNE

I Oblique (Slant) Asymptote

The line $y=ax+b$ is an oblique (slant) asymptote for the curve $y=f(x)$ if:

$$\lim_{x\to\pm\infty}[f(x)-(ax+b)]=0$$

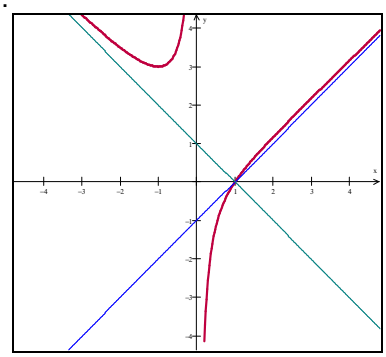
Notes:

1. An oblique asymptote may be crossed or touched by the graph of the function
2. The graph of a function may have at most two oblique asymptotes (one as $x\to-\infty$ and one as $x\to\infty$)

as $x\to-\infty$ $\quad y=-x+1$

as $x\to+\infty$ $\quad y=x-1$

Ex 8. Find the equations of the oblique asymptotes for the function represented below (oblique asymptotes are also in the figure).



J Oblique Asymptotes for Rational Functions

A rational function of the form:

$$f(x)=\frac{P_n(x)}{Q_m(x)}=\frac{a_nx^n+a_{n-1}x^{n-1}+\ldots+a_2x^2+a_1x+a_0}{b_mx^m+b_{m-1}x^{m-1}+\ldots+b_2x^2+b_1x+b_0}$$

has an oblique (slant) asymptote if $n=m+1$.

Note. To get the equation of the oblique (slant) asymptote, use the long division algorithm to write the rational function in the form:

$$f(x)=\frac{P_n(x)}{Q_m(x)}=ax+b+\frac{R(x)}{Q_m(x)}$$

where $0\le degree(R)<degree(Q_m)=m$

Finally, the equation of the oblique (slant) asymptote is given by:

$$y=ax+b$$

Ex 9. Consider the rational function $y=f(x)=\frac{x^2}{x-1}$.

Find the equation of the oblique asymptote.

$$\begin{array}{r|l} & x+1 \\ \hline x-1 & x^2 \\ & x^2-x \\ \hline & 1+x \\ & x-1 \\ \hline & 1 \end{array}$$

$$\frac{x^2}{x-1}=x+1+\frac{1}{x-1}$$

$\therefore y=x+1$ is the oblique asymptote

Reading: Nelson Textbook, Pages 181-192

Homework: Nelson Textbook: Page 193 #1ab, 3bd, 4de, 5cd, 7ac, 9bd, 14

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